DP Biology – A1.1 (HL extension)

Where water came from

- Which features of water make life possible?
- What does living in water make easier and harder for organisms?

HL statements A1.1.7–8. Textbook pp. 11–13.
Image: the Earth from Apollo 17. NASA, public domain.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	A1.1.7–8	Evaluate the asteroid hypothesis for the origin of Earth's water, explain why Earth kept it, and explain why the search for life beyond Earth starts with water

A wet planet that formed dry

A1.1.7



Earth from Apollo 17. NASA, public domain.

The Earth formed about **4.5 billion years ago** from rock and dust close to the Sun. There it was too hot for water vapour to condense, so the young Earth was probably dry.

Further from the Sun, beyond the "snow line", water froze into ice on asteroids and comets. So most of Earth's water probably arrived from **outside the planet**: it is **extraplanetary**.

Water has been present for at least 4 billion years, long enough for life to arise and evolve in it.

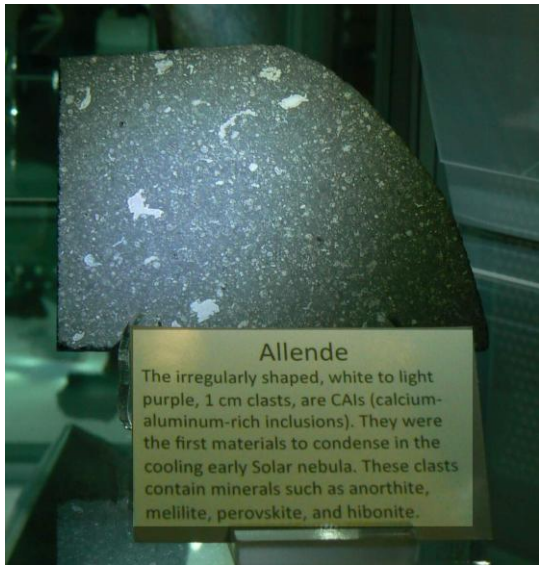
Question

If the young Earth was too hot to hold water, how did it end up with oceans covering 70% of its surface?

The asteroid hypothesis



The asteroid Vesta. NASA/JPL, public domain.



Allende, a carbonaceous chondrite. Milo44, CC BY-SA 3.0.

The leading hypothesis is that water was delivered by **asteroids**, especially the kind that formed further from the Sun, during the early bombardment of the young Earth.

Fragments of these asteroids still fall as **carbonaceous chondrite** meteorites, such as Allende.

Evidence	What it shows
Water in meteorites	Some carbonaceous chondrites hold up to about a fifth of their mass as water, locked in minerals
Isotope ratio	The ratio of deuterium to hydrogen in their water matches Earth's oceans (next slide)

Evidence from deuterium

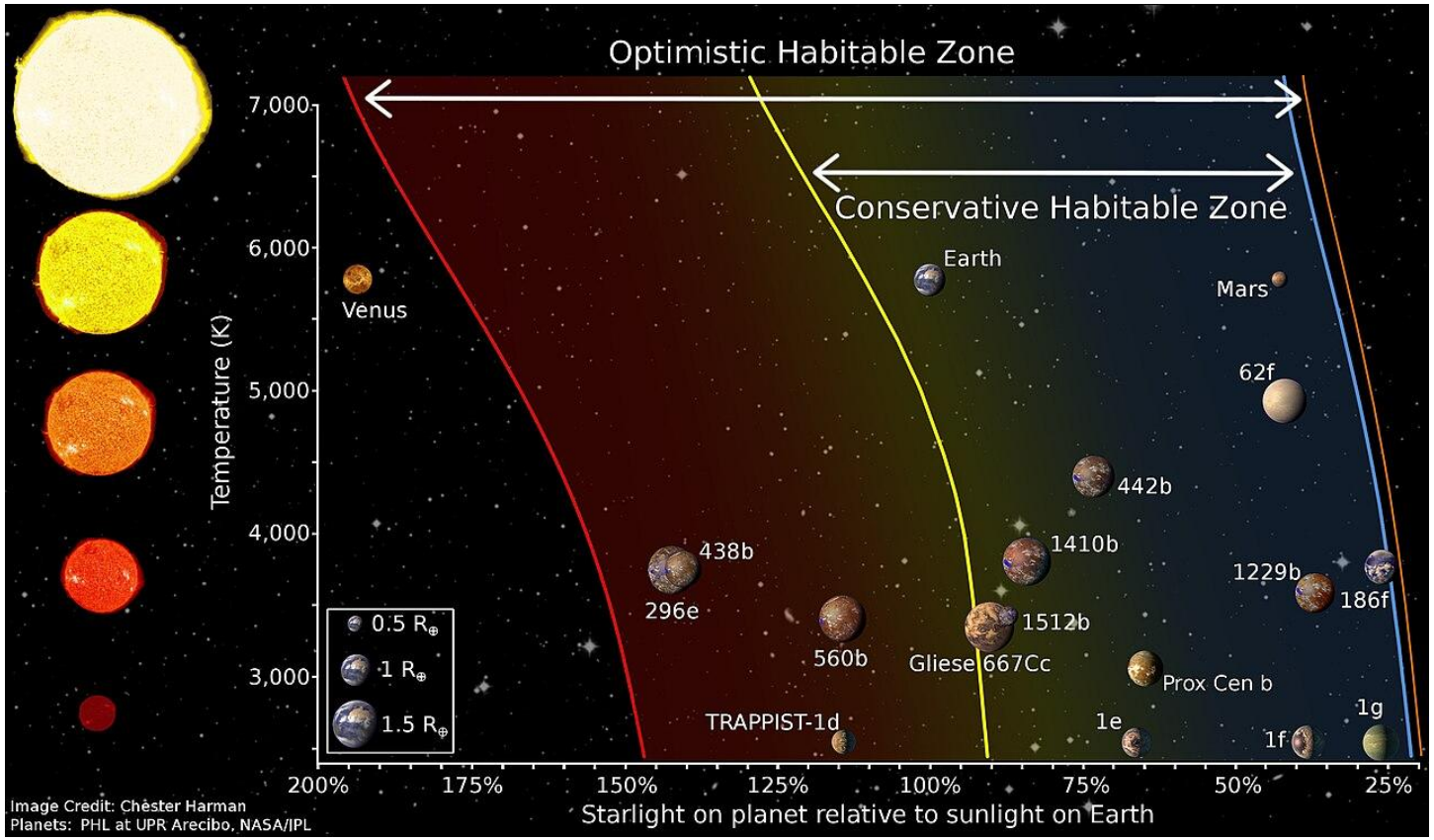
A1.1.7

Deuterium is hydrogen with one neutron. Water that formed in different parts of the solar system has different ratios of deuterium to ordinary hydrogen (D/H), so the ratio works like a fingerprint for where water came from.

Source of water	D/H ratio (parts per 10 000)
Earth's oceans	1.56
Carbonaceous chondrite meteorites	about 1.4 to 1.8
Most comets (such as Halley)	about 3
Comet 67P (Rosetta, 2014)	5.3
Comet Hartley 2	1.6

1. Which source matches ocean water best?
2. What does comet 67P suggest?
3. Why is one comet (Hartley 2) not enough to support a comet origin?

The habitable (Goldilocks) zone



Habitable zones around stars of different temperatures. Chester Harman, CC BY-SA 4.0.

The **habitable zone** is the range of distances from a star where a planet's surface could hold **liquid water**: not so close that it boils away, not so far that it freezes.

Cooler stars have habitable zones close in; hotter stars have them further out. Earth lies inside the Sun's.

Life as we know it needs liquid water as a solvent and a medium for reactions, so searching for water narrows the search for life.

Why Earth kept its water

A1.1.7

	Venus	Earth	Mars
Distance from Sun	Closer	In the habitable zone	Further
Size and gravity	Similar to Earth	Strong enough to hold an atmosphere	Smaller: weaker gravity
Surface temperature	About 460 °C	Mostly between 0 and 100 °C	Mostly below 0 °C
Water now	Lost to space	Liquid oceans	Frozen or underground

Key idea

Earth kept its water because its **gravity** holds an atmosphere and water vapour, and its **temperature** is low enough for water vapour to condense into liquid.

How scientists look for water beyond Earth

A1.1.8

Transit spectroscopy: when a planet passes in front of its star, starlight passes through the planet's atmosphere. Water molecules absorb particular wavelengths, leaving a signature in the spectrum that telescopes can detect.

Closer to home, probes and rovers have found evidence of water on **Mars**, and of liquid oceans under the ice of moons such as **Europa** (Jupiter) and **Enceladus** (Saturn), where water spurts out in plumes.

Discuss

A planet in the habitable zone might still have no life. Suggest reasons, thinking about size, atmosphere, magnetic field and the time needed for life to arise.

Check-in: water beyond Earth

A1.1.7–8

1. Explain why the young Earth probably had little water of its own.
2. Outline two pieces of evidence that asteroids delivered Earth’s water.
3. State two reasons why Earth kept its water.
4. Define the habitable (Goldilocks) zone.
5. Explain why the search for extraterrestrial life focuses on water.

How to use this

Answer without your notes, then compare with a partner.

Common mistakes to avoid

Not this	But this
Earth's water mostly came from comets.	The D/H evidence points mainly to asteroids; most comets do not match.
Any planet in the habitable zone has life.	The zone only means liquid water is possible; many other conditions matter.
Mars never had water.	Mars once had liquid water; it is now frozen or underground.
Earth keeps its water only because of its temperature.	Gravity (to hold the atmosphere) and temperature both matter.

Check that you can...

1. **Explain** why the young Earth could not condense its own water
2. **Evaluate** the evidence that asteroids delivered Earth's water, using D/H ratios
3. **Explain** how gravity and temperature let Earth keep its water
4. **Define** the habitable zone
5. **Outline** how scientists search for water beyond Earth
6. **Explain** why the search for life starts with water

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Give one piece of evidence that asteroids brought water to Earth.
2. State two reasons Earth has kept its water.
3. Explain why the search for life beyond Earth focuses on water.

Image credits

The Earth seen from Apollo 17.jpg – NASA/Apollo 17 crew; taken by either Harrison Schmitt or Ron – Public domain

Vesta in natural color (cropped).jpg – NASA / JPL / MPS / DLR / IDA / Björn Jónsson – Public domain

Allende meteorite slice ASU.jpg – Milo44 – CC BY-SA 3.0

Diagram of different habitable zone regions by Chester Harman.jpg – Chester Harman – CC BY-SA 4.0

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